

# Revision · Transformations

Cambridge insert: reflection, rotation and translation — three ways to move a shape without changing it.

LESSON 62–63

2 × 40 MIN

GEOMETRY 8 – REVISION

STAGE 9 · 13.3

LESSON CLOCK

6'

12'

12'

6'

4'

|                                       |        |
|---------------------------------------|--------|
| Fold, turn and slide a paper triangle | 6 min  |
| Reflection                            | 12 min |
| Rotation                              | 12 min |
| Translation and combinations          | 6 min  |
| Homework                              | 4 min  |

LEARNING OBJECTIVES

- Reflect a shape in a given mirror line.
- Rotate a shape about a given centre through a given angle.
- Translate a shape by a given vector.
- Describe a transformation fully, and combine two of them.

TEXTBOOK

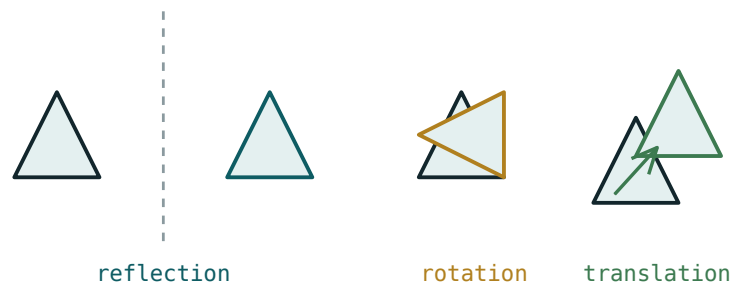
|           |                            |
|-----------|----------------------------|
| National  | Annual revision            |
| Cambridge | Learner’s Book pp. 297–306 |
| Workbook  | Workbook 13.3              |

01

## Explanation

### The three isometries

All three of these move a shape without changing its size or its shape. The image is always **congruent** to the object — so they are called **isometries**.



*Reflection flips, rotation turns, translation slides — in every case the image is congruent to the object.*

| TRANSFORMATION | DESCRIBED FULLY BY          | POINTS THAT DO NOT MOVE        |
|----------------|-----------------------------|--------------------------------|
| Reflection     | the mirror line             | every point on the mirror line |
| Rotation       | centre, angle and direction | the centre only                |
| Translation    | a vector                    | none                           |

**“DESCRIBE FULLY” IS A MARK SCHEME**

“A rotation” earns nothing. “A rotation of  $90^\circ$  clockwise about  $(0; 0)$ ” earns full marks. Name the transformation and every piece of data it needs.

## Doing each one

**Reflection.** Each point moves perpendicular to the mirror, to the same distance on the other side. In coordinates:

in the  $x$ -axis:  $(x; y) \rightarrow (x; -y)$  | in the  $y$ -axis:  $(x; y) \rightarrow (-x; y)$

in the line  $y = x$ :  $(x; y) \rightarrow (y; x)$

**Rotation.** Every point turns through the same angle about the centre, staying the same distance from it. About the origin:

$90^\circ$  anticlockwise:  $(x; y) \rightarrow (-y; x)$  |  $180^\circ$ :  $(x; y) \rightarrow (-x; -y)$

**Translation.** Every point moves by the same vector:

$(x; y) \rightarrow (x + a; y + b)$  for the vector  $(a; b)$

Two translations combine into one — just add the vectors. Two reflections in **parallel** mirrors give a translation; two reflections in **intersecting** mirrors give a rotation about the crossing point.

## 02 Worked examples

**EXAMPLE 1** Reflect  $A(3; 2)$  in the  $x$ -axis, then in the  $y$ -axis. Describe the single transformation that has the same effect.

①  $(3; 2) \rightarrow (3; -2)$

Reflection in the  $x$ -axis.

②  $(3; -2) \rightarrow (-3; -2)$

Reflection in the  $y$ -axis.

③  $(3; 2) \rightarrow (-3; -2)$

Both coordinates have changed sign.

④ A rotation of  $180^\circ$  about the origin.

Two perpendicular mirrors give a half-turn about their crossing point.

ANSWER

A rotation of  $180^\circ$  about  $(0; 0)$

**EXAMPLE 2** Rotate  $B(4; 1)$  through  $90^\circ$  anticlockwise about the origin, then translate by  $(-2; 3)$ .

①  $(4; 1) \rightarrow (-1; 4)$

The rule  $(x; y) \rightarrow (-y; x)$ .

②  $(-1; 4) \rightarrow (-3; 7)$

Add the translation vector.

ANSWER

$(-3; 7)$

**EXAMPLE 3** Describe fully the transformation taking  $(2; 5)$  to  $(5; 2)$  and  $(1; 7)$  to  $(7; 1)$ .

① In both cases the coordinates swap.

②  $(x; y) \rightarrow (y; x)$

That is the rule for one particular mirror.

③ A reflection in the line  $y = x$ .

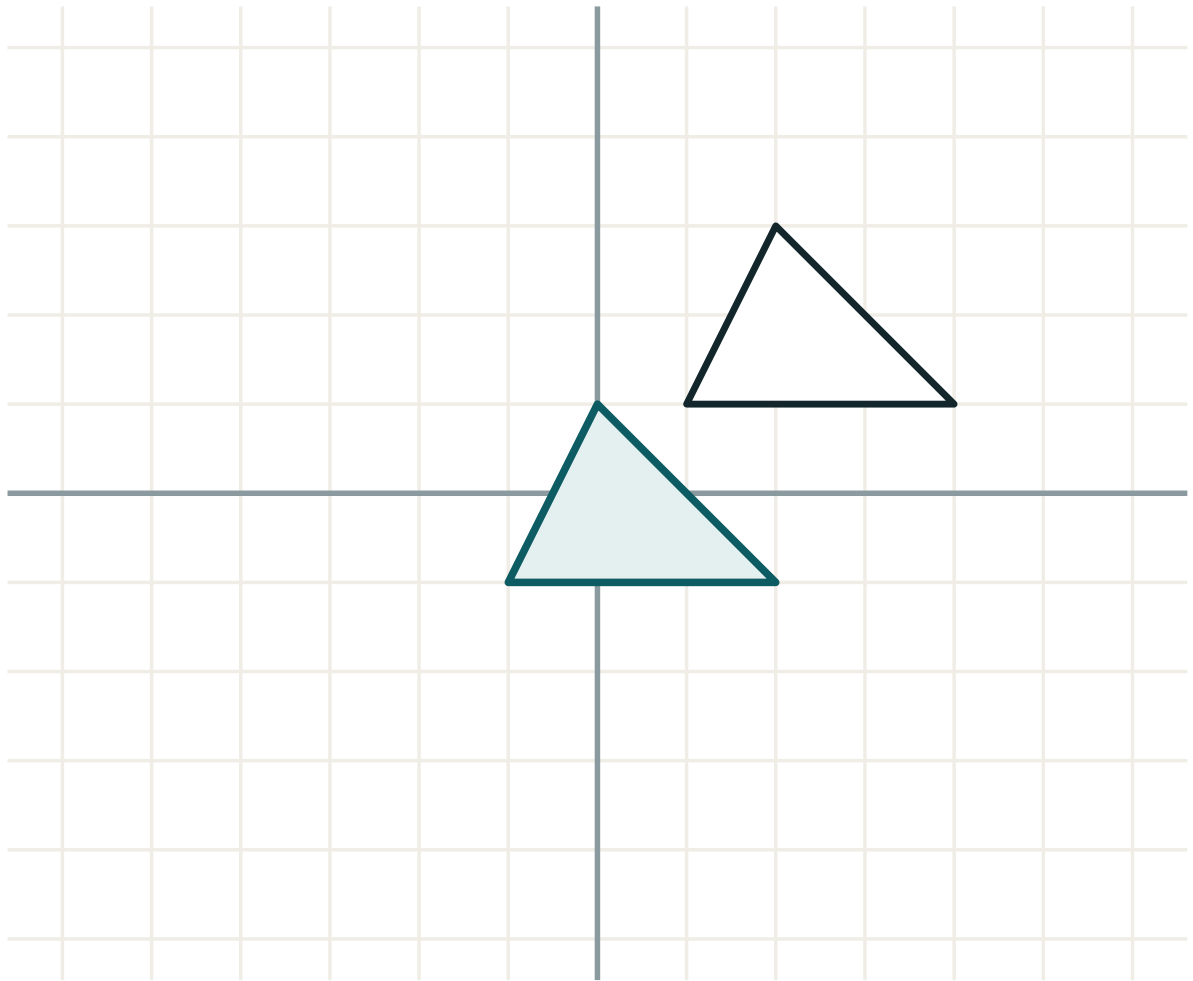
Check: the midpoint of each pair lies on  $y = x$  ✓

ANSWER

A reflection in  $y = x$

### 03 Interactive model

Choose a transformation and watch the shape move; compare object and image.

**Reflect, rotate, translate**transformation **translation**lengths **unchanged**angles **unchanged**orientation **kept****04 Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH            | O'ZBEKCHA                 | РУССКИЙ               |
|--------------------|---------------------------|-----------------------|
| Transformation     | Almashtirish              | Преобразование        |
| Reflection         | Simmetriya (aks ettirish) | Отражение (симметрия) |
| Mirror line        | Simmetriya o'qi           | Ось симметрии         |
| Rotation           | Burish                    | Поворот               |
| Centre of rotation | Burilish markazi          | Центр поворота        |
| Translation        | Parallel ko'chirish       | Параллельный перенос  |
| Image              | Tasvir                    | Образ                 |
| Object             | Asl figura                | Прообраз              |
| Congruent          | Teng (mos)                | Равные (конгруэнтные) |
| Invariant point    | O'zgarmas nuqta           | Неподвижная точка     |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

Reflecting  $(4; -3)$  in the  $x$ -axis gives:

A rotation of  $180^\circ$  about the origin takes  $(2; 5)$  to:

Translating  $(1; 1)$  by  $(3; -4)$  gives:

Which transformation has no invariant point?

reflection

rotation

translation

all have one

To describe a rotation fully you need:

the angle only

the centre only

centre, angle and direction

a vector

## 06 Practice · 21 problems

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Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

**Easy · warm the idea up**

7 PROBLEMS



1. *Reflect  $(5; 2)$  in the  $x$ -axis.*

ANSWER  $(5; -2)$

2. *Reflect  $(5; 2)$  in the  $y$ -axis.*

ANSWER  $(-5; 2)$

3. *Reflect  $(5; 2)$  in the line  $y = x$ .*

ANSWER  $(2; 5)$

4. *Rotate  $(3; 0)$  through  $180^\circ$  about the origin.*

ANSWER  $(-3; 0)$

5. *Translate  $(2; 3)$  by  $(4; 1)$ .*

ANSWER  $(6; 4)$

6. *Rotate  $(1; 0)$  through  $90^\circ$  anticlockwise about the origin.*

ANSWER  $(0; 1)$

7. *Translate  $(-1; 5)$  by  $(2; -5)$ .*

ANSWER  $(1; 0)$

**Medium · the standard question**

7 PROBLEMS

1. Rotate  $(4; 2)$  through  $90^\circ$  anticlockwise about the origin.

ANSWER  $(-2; 4)$

2. Rotate  $(4; 2)$  through  $90^\circ$  clockwise about the origin.

ANSWER  $(2; -4)$

3. Reflect  $(-3; 6)$  in the line  $y = x$ .

ANSWER  $(6; -3)$

4. Describe fully the transformation taking  $(1; 2)$  to  $(1; -2)$ .

ANSWER A reflection in the  $x$ -axis.

5. Describe fully the transformation taking  $(2; 3)$  to  $(7; 8)$ .

ANSWER A translation by  $(5; 5)$ .

6. Reflect  $(2; 5)$  in the  $x$ -axis, then in the  $y$ -axis.

ANSWER  $(-2; -5)$

7. Translate  $(0; 0)$  by  $(3; 4)$ , then by  $(-1; 2)$ .

ANSWER  $(2; 6)$

### Hard · stretch and reason

7 PROBLEMS

1. Describe fully the single transformation equal to a reflection in the  $x$ -axis followed by a reflection in the  $y$ -axis.

ANSWER A rotation of  $180^\circ$  about the origin.

2. Rotate the triangle  $(0;0)$ ,  $(3;0)$ ,  $(0;2)$  through  $90^\circ$  anticlockwise about the origin.

ANSWER  $(0;0)$ ,  $(0;3)$ ,  $(-2;0)$

3. Describe fully the transformation taking  $(1; 3)$  to  $(3; 1)$  and  $(4; 6)$  to  $(6; 4)$ .

ANSWER A reflection in the line  $y = x$ .

4. A shape is translated by  $(3; 1)$  and then by  $(-3; -1)$ . Describe the result.

ANSWER The identity — every point returns to where it started.

5. Rotate  $(5; 2)$  through  $180^\circ$  about the point  $(1; 1)$ .

ANSWER  $(-3; 0)$  — the centre is the midpoint of object and image

6. Two parallel mirrors are 4 units apart. Describe the result of reflecting in one, then the other.

ANSWER A translation of 8 units, perpendicular to the mirrors — twice the distance between them.

7. Explain why a reflection reverses orientation but a rotation does not.

ANSWER A rotation can be carried out by sliding the shape within the plane, so its sense (clockwise round the vertices) is preserved; a reflection requires flipping it over, which reverses that sense.

## 07 Homework

### Homework — 6 problems

Cambridge Stage 9, Unit 13.3. Describe every transformation fully.

1. Reflect  $(6; -2)$  in the  $x$ -axis, the  $y$ -axis and the line  $y = x$ .
2. Rotate  $(3; 5)$  through  $90^\circ$  anticlockwise about the origin.

3. *Translate  $(-2; 4)$  by  $(5; -6)$ .*
4. *Describe fully the transformation taking  $(2; 7)$  to  $(-2; 7)$ .*
5. *Rotate the triangle  $(0;0), (4;0), (0;3)$  through  $180^\circ$  about the origin.*
6. In one sentence, say what all three isometries have in common.